FRIDAY AUGUST 31st 2018 LECTURE 4

Define: $\vec{L}$ = LinMom =

$\vec{H}_{/o}$ = Angular Momentum about (ω, r, t) $o \equiv \vec{r} \times m\vec{v}$

E_k = Kinetic energy $= \frac{1}{2}mv^2 = \frac{1}{2}m\vec{v}\cdot\vec{v}$

Note: given $\vec{F} = m\vec{a}$ & I.C. $(\vec{r}(0) = \vec{r}_0, \vec{v}(0) = \vec{v}_0)$

$\rightarrow$ Motion (computer or analytical solution)

Depending on features of $\vec{F}(\vec{r}, \vec{v}, t)$ solution has various properties

Linear Momentum

$$\vec{F} = m\vec{a}$$

$$\int \vec{F}\,dt = \int m\vec{a}\,dt$$

$$\int_{t_1}^{t_2} \vec{F}\,dt = m(\vec{v}_2 - \vec{v}_1) = m\Delta\vec{v}$$

finite change

$$\boxed{\vec{P} = \Delta\vec{L}}$$

impulse

change in linear momentum

$\Rightarrow$ LINEAR MOMENTUM PRINCIPAL

Special case:

- $\vec{F} \cdot \hat{\lambda} = 0$

  constant

  $\Rightarrow m(\Delta\vec{v}) \cdot \hat{\lambda} = 0$

  $\Rightarrow \hat{\lambda} = \hat{\imath} \Rightarrow V_x = \text{constant}$

- $\vec{F} = \vec{0}$

  $\Rightarrow \vec{v}$ constant, $\vec{L}$ constant

$\equiv$ conservation of linear momentum ex) Ballistics w/no friction

$\vec{F} = -mg\hat{j}$

$\vec{F} \cdot \hat{i} = 0$

$\Rightarrow V_x \equiv \vec{v} \cdot \hat{i} = \text{constant}$

How to use:

$\boxed{\text{impulse } \vec{F}}$ add to list of ODE's

Imp

Angular Momentum

Generally can be w.r.t. any point C, for now, fixed pt. O

$\vec{F} = m\vec{a}$

$\vec{r} \times \vec{F} = \vec{r} \times (m\vec{a})$

$= m(\vec{r} \times \dot{\vec{v}})$

Note. $\frac{d}{dt}(\vec{r} \times \vec{v}) = \underbrace{\cancel{\dot{\vec{r}} \times \vec{v}}}_{\substack{\dot{\vec{r}} = \vec{v} \\ \text{vector} \times \text{vector} \\ = 0}} + \vec{r} \times \dot{\vec{v}} = \vec{r} \times \dot{\vec{v}} = \vec{r} \times \vec{a}$

$\Rightarrow \vec{M}_{/0} = \underbrace{\dot{\vec{H}}_{/0}}_{\text{angular momentum}}$

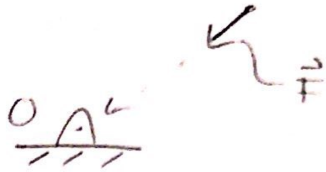
$\Rightarrow \underbrace{\text{Angular impulse}}_{\int \vec{r} \times \vec{F} dt} = \Delta\vec{H}_0$

Conservation of angular momentum:

If $\vec{r} \times \vec{F} = 0 \longrightarrow \vec{H}_{/0} = \text{constant}$

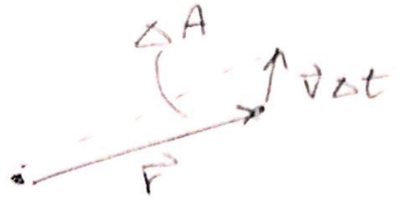
$\Delta\vec{H}_{/0} = \vec{0}$

ex| central force



$\vec{F} = \text{something} \cdot \hat{e}_r$ $\quad \hat{e}_r = \frac{\vec{r}}{|\vec{r}|}$

Equal areas in equal times



$$\Delta A = |\vec{r}||\vec{v}\Delta t|\sin\theta$$
$$= |\vec{r} \times \vec{v}|\Delta t$$
$$\dot{A} = |\vec{r} \times \vec{v}|$$

Note: does not depend on inverse square law

Energy:

$$\vec{F} = m\vec{a}$$
$$\vec{v} \cdot \vec{F} = \underbrace{m\vec{a} \cdot \vec{v}}$$

aside: $\frac{d}{dt}(v^2) = 2v\dot{v}$

$$\frac{d}{dt}(\vec{v} \cdot \vec{v}) = 2\vec{v} \cdot \dot{\vec{v}} = 2\vec{v} \cdot \vec{a}$$

$$= \frac{d}{dt}\frac{1}{2}m(\vec{v} \cdot \vec{v})$$

$$\boxed{P = \frac{d}{dt}E_k} \qquad \boxed{P = \dot{E}_k}$$

power

If $\vec{F} \perp \vec{v}$ always $\Rightarrow \boxed{E_k = \text{constant}}$

$$\Rightarrow \int P\,dt = \Delta E_k = E_{k_2} - E_{k_1}$$

$$\int P\,dt = \int \vec{F} \cdot \vec{v}\,dt = \int \vec{F} \cdot \frac{d\vec{r}}{dt}\,dt = \int \vec{F} \cdot d\vec{r} = \text{Work}$$

$$\int_{t_1}^{t_2} P\,dt = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r} = \text{Work} = \Delta E_k$$

Add to ODEs, $\dot{W} = \vec{F} \cdot \vec{v}$